

EE/CE 252/251

Signals and Systems

with

Dr. Naveed R. Butt

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Habib University

Problem

Consider the system defined by $y(t) = S\{x(t)\} = x(t) \cos(\omega t) + x(t - 1) \sin(\omega t)$, where ω is a non-zero constant. Determine whether this system is **(a) Memoryless (b) Causal (c) Linear (d) Time-Invariant, (e) BIBO Stable**. Provide proper justifications (w.r.t given system) for your answers.

Solution

- (a) Not memoryless, since output depends on a past value also.
- (b) Causal, since output does not require any future input.
- (c) For linearity we have to show that

$$S\{\alpha_1 x_1(t) + \alpha_2 x_2(t)\} = \alpha_1 S\{x_1(t)\} + \alpha_2 S\{x_2(t)\}$$

Let's find $S\{x_1(t)\}$, $S\{x_2(t)\}$, and $S\{\alpha_1 x_1(t) + \alpha_2 x_2(t)\}$ and plug in the above to see if the equality holds. We have

$$S\{x_1(t)\} = x_1(t) \cos(\omega t) + x_1(t-1) \sin(\omega t)$$

$$S\{x_2(t)\} = x_2(t) \cos(\omega t) + x_2(t-1) \sin(\omega t)$$

And

$$S\{\alpha_1 x_1(t) + \alpha_2 x_2(t)\} = (\alpha_1 x_1(t) + \alpha_2 x_2(t)) \cos(\omega t) + (\alpha_1 x_1(t-1) + \alpha_2 x_2(t-1)) \sin(\omega t)$$

Plugging the above three in the test condition shows that it holds. Therefore, system is linear

Solution

(d) For time-invariance, we have to show that if $y(t) = S\{x(t)\}$ and $y_2(t) = S\{x(t - t_0)\}$, then $y_2(t) = y(t - t_0)$. Let's check:

$$y(t) = S\{x(t)\} = x(t) \cos(\omega t) + x(t - 1) \sin(\omega t)$$

$$y_2(t) = S\{x(t - t_0)\} = x(t - t_0) \cos(\omega t) + x(t - 1 - t_0) \sin(\omega t),$$

And

$$y(t - t_0) = x(t - t_0) \cos(\omega t - \omega t_0) + x(t - 1 - t_0) \sin(\omega t - \omega t_0) \neq y_2(t)$$

Which leads to the conclusion that the given system is not time-invariant.

Solution

(e) We start by noticing that

$$\begin{aligned} |y(t)| &= |x(t) \cos(\omega t) + x(t-1) \sin(\omega t)| \\ &\leq |x(t) \cos(\omega t)| + |x(t-1) \sin(\omega t)| \quad (\text{since } |a+b| \leq |a| + |b|) \\ &\leq |x(t)| + |x(t-1)| \quad (\text{since } |\cos(\omega t)| \leq 1 \text{ and } |\sin(\omega t)| \leq 1) \end{aligned}$$

Which leads to the conclusion that the output is bounded as long as the input is bounded. Thus the system is BIBO stable.

Problem

Consider a DT system described as $y[n + 1] = x[n + 1]x[n]$. Determine whether the system is **(a)** linear, **(b)** time-invariant, **(c)** causal, **(d)** invertible, **(e)** BIBO-stable, and **(f)** memoryless.

Solution

Let us delay the input–output equation by one to obtain the equivalent but more convenient representation of $y[n] = x[n]x[n - 1]$.

(a) Linearity requires both homogeneity and additivity. Let us first investigate homogeneity. Assuming $x[n] \implies y[n]$, we see that

$$ax[n] \implies (ax[n])(ax[n - 1]) = a^2y[n] \neq ay[n]$$

Thus, the system does not satisfy the homogeneity property.

The system also does not satisfy the additivity property. Assuming $x_1[n] \implies y_1[n]$ and $x_2[n] \implies y_2[n]$, we see that input $x[n] = x_1[n] + x_2[n]$ produces output $y[n]$ as

$$\begin{aligned} y[n] &= (x_1[n] + x_2[n])(x_1[n - 1] + x_2[n - 1]) \\ &= x_1[n]x_1[n - 1] + x_2[n]x_2[n - 1] + x_1[n]x_2[n - 1] + x_2[n]x_1[n - 1] \\ &= y_1[n] + y_2[n] + x_1[n]x_2[n - 1] + x_2[n]x_1[n - 1] \\ &\neq y_1[n] + y_2[n] \end{aligned}$$

Clearly, additivity is not satisfied.

Since the system does not satisfy both the homogeneity and additivity properties, we conclude that the system is not linear.

Homogeneity and additivity both tested here for practice. Otherwise, we know that even if one of the two fails, the system is nonlinear.

(b) To be time-invariant, a shift in any input should cause a corresponding shift in respective output. Assume that $x[n] \implies y[n]$. Applying a delay version of this input to the system yields

$$x[n - N] \implies x[n - N]x[n - 1 - N] = x[(n - N)]x[(n - N) - 1] = y[n - N]$$

Since shifting an input causes a corresponding shift in the output, we conclude that the system is time-invariant.

(c) To be causal, an output value cannot depend on any future input values. The output y at time n depends on the input x at present and past times n and $n - 1$. Since the current output does not depend on future input values, the system is causal.

(d) For a system to be invertible, every input must generate a unique output, which allows exact recovery of the input from the output. Consider two inputs to this system: $x_1[n] = 1$ and $x_2[n] = -1$. Both inputs generate the same output: $y_1[n] = y_2[n] = 1$. Since unique inputs do not always generate unique outputs, we conclude that the system is not invertible.

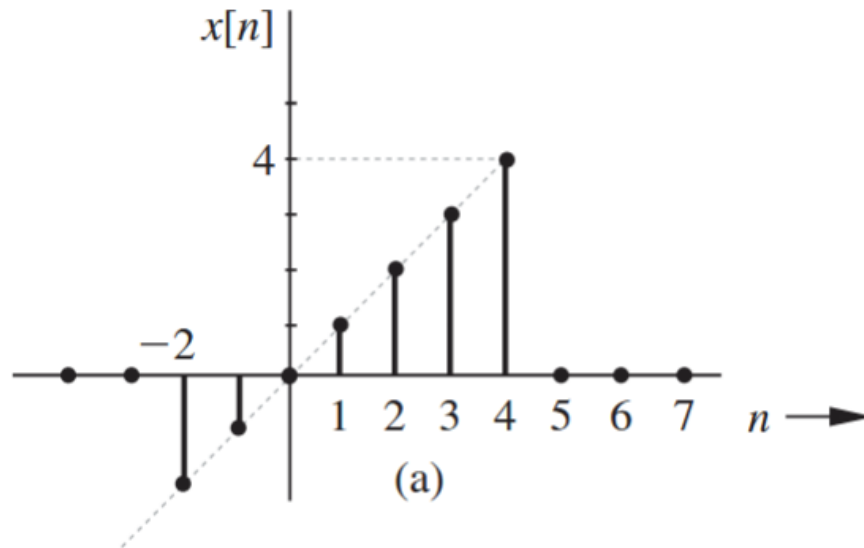
(e) To be BIBO-stable, any bounded input must generate a bounded output. A bounded input satisfies $|x[n]| \leq M_x < \infty$ for all n . Given this condition, the system output magnitude behaves as

$$|y[n]| = |x[n]x[n-1]| = |x[n]| |x[n-1]| \leq M_x^2 < \infty$$

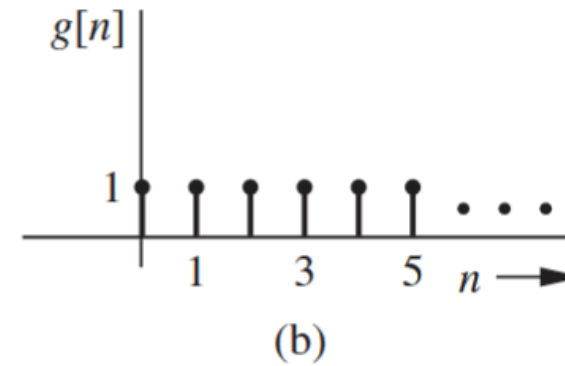
Since any bounded input is guaranteed to produce a bounded output, it follows that the system is BIBO-stable.

(f) To be memoryless, a system's output can only depend on the strength of the current input. Since the output y at time n depends on the input x not only at present time n but also on past time $n-1$, we see that the system is not memoryless.

Problem



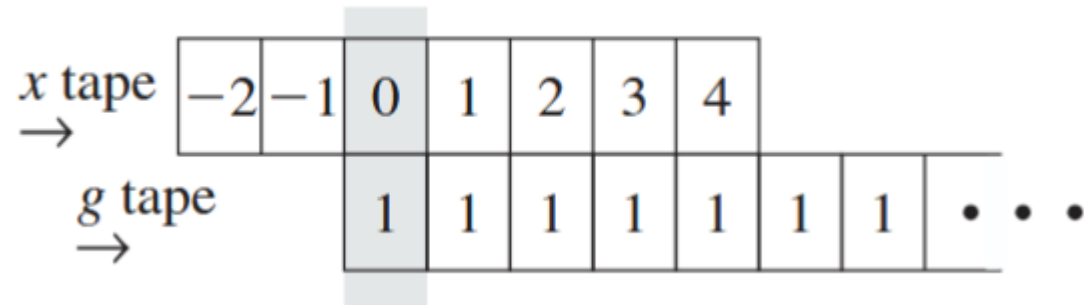
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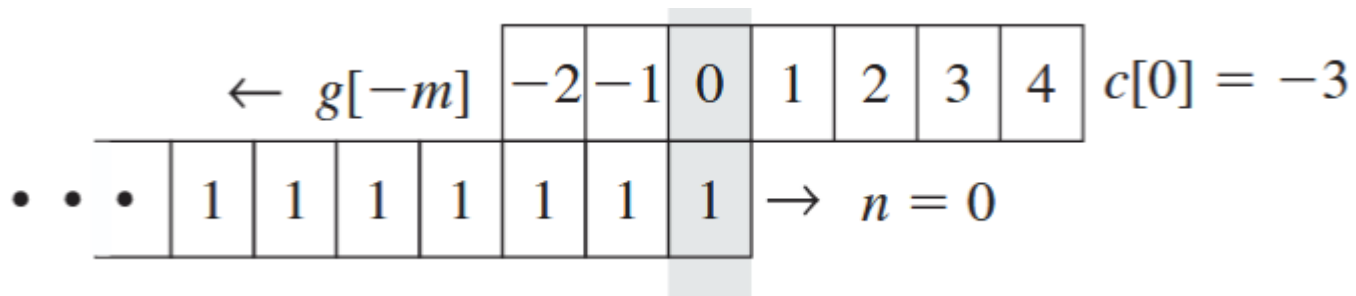
Use sliding-tape method to find $c[n] = x[n] * g[n]$

Solution

Step 1: Write the two sequences on tapes such that their zeros align (insert zeros in a given function if required)

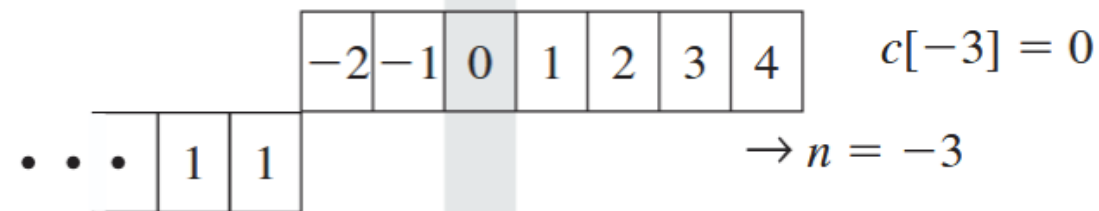
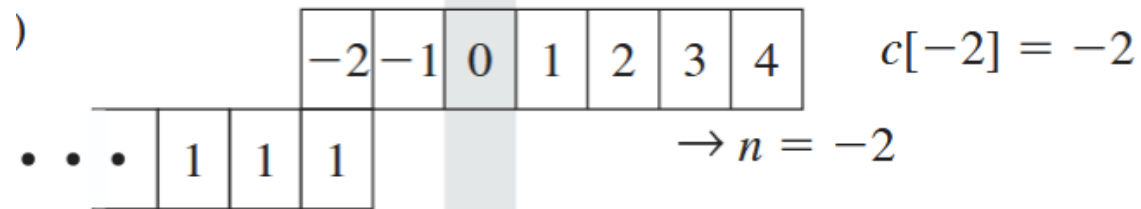
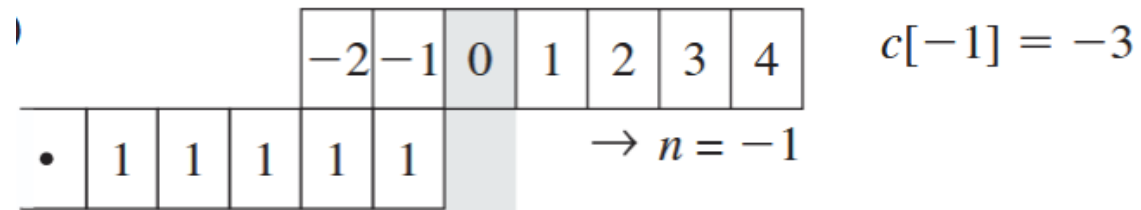
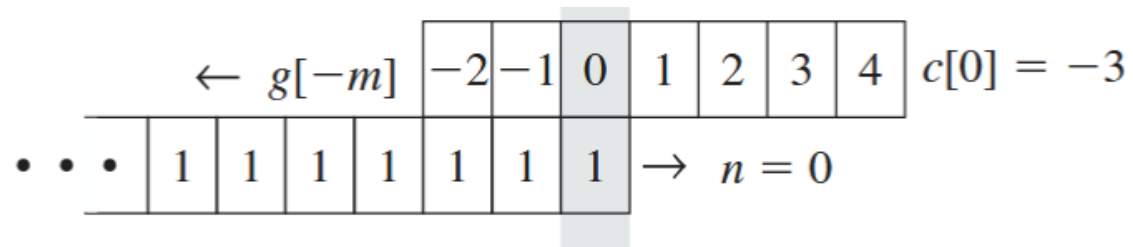


Step 2: Flip any one of them (since convolution is commutative) about the aligned axis.

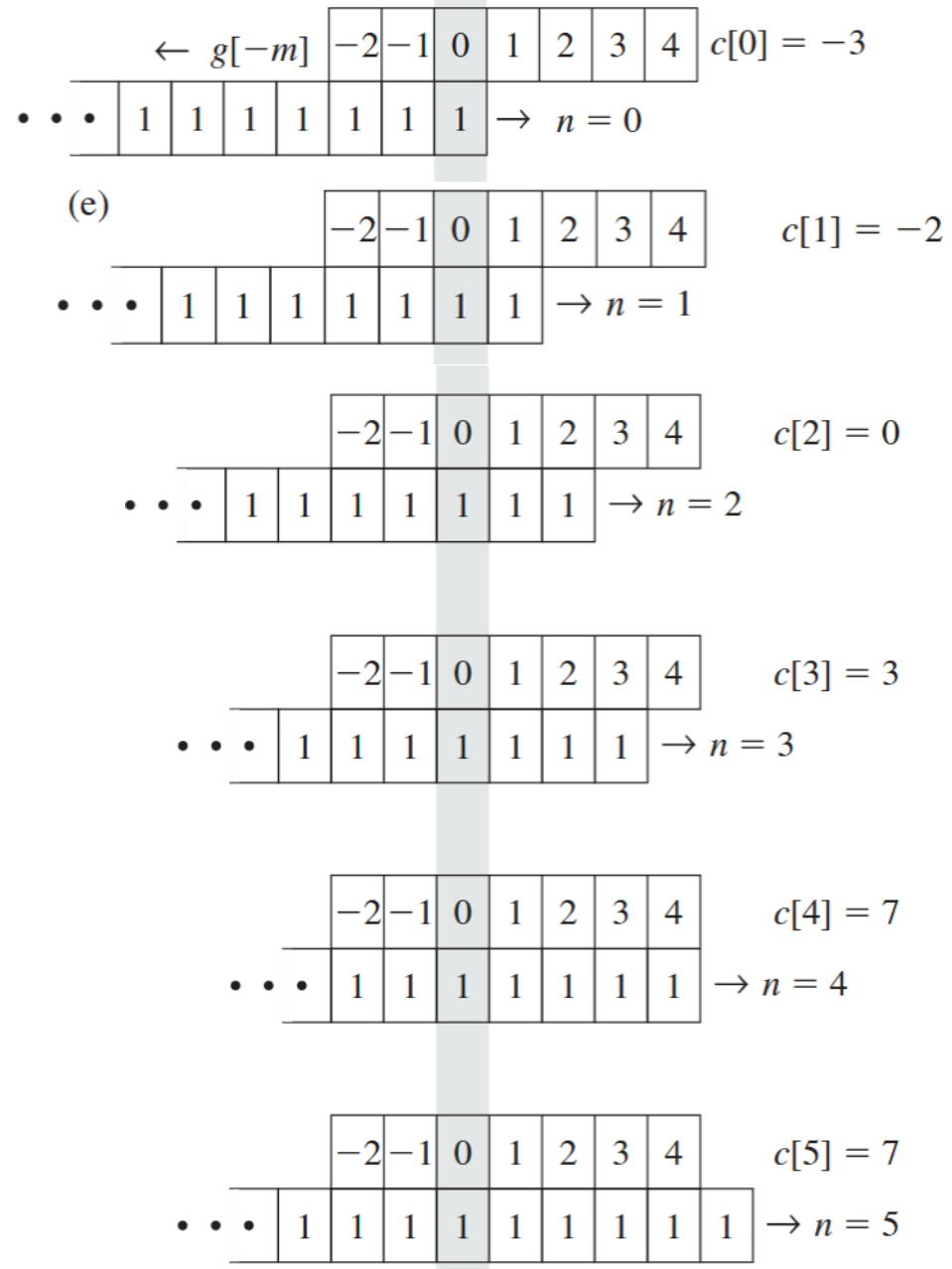


Step 3: Get $c[0]$ by multiplying overlapping cells and adding the result.

Step 4: Slide flipped one to the right by one step and repeat to get $c[1]$ and slide to left by one and repeat to get $c[-1]$ etc.



Sliding left will give
 $c[-1], c[-2], c[-3] \dots$



Sliding right will give
 $c[1], c[2], c[3] \dots$

Problem

For a system specified by equation

$$y[n] = x[n] - 2x[n - 1]$$

Find the system response to input $x[n] = u[n]$.

Solution

- We first note that the given system can be easily shown to be LTI. Therefore output of this system to any input is simply the convolution
 - $y[n] = x[n] * h[n]$
 - Where $x[n]$ is the input, and $h[n]$ the impulse response.
- Next, we find the system impulse response by setting $x[n] = \delta[n]$
 - $y[n] = h[n] = \delta[n] - 2\delta[n - 1]$

Solution

- Finally, we find the output of the system to the given input $x[n] = u[n]$

$$\begin{aligned} y[n] &= x[n] * h[n] = u[n] * (\delta[n] - 2\delta[n - 1]) = u[n] * \delta[n] - 2(u[n] * \delta[n - 1]) \\ &= u[n] - 2u[n - 1] \end{aligned}$$

Note: in the above we have used

- Distributive property of convolution
- Fact that $\delta(t)$ is the identity of convolution
- Shifting property of convolution

Problem

Consider the DT LTI system with input $x[n]$ and output $y[n]$

$$y[n] = x[n] + \frac{1}{2}y[n-1]$$

Assume that the impulse response of the system satisfies $h[k] = 0, k < 0$.

- (a) Find values of $h[0]$, $h[1]$, and $h[2]$
- (b) Find a general simplified formulation of $h[n]$
- (c) Use $h[n]$ to check whether the system is causal.
- (d) Use $h[n]$ to check whether the system is memoryless.
- (e) Use $h[n]$ to check whether the system is BIBO stable.

Justify your answers mathematically.

Solution

Let the input be an impulse $x[n] = \delta[n]$, then the output is impulse response $y[n] = h[n]$, leading to

$$h[n] = \delta[n] + \frac{1}{2}h[n-1]$$

(a) From the above, we get

$$h[0] = \delta[0] + \frac{1}{2}h[-1] = \delta[0] = 1$$

$$h[1] = \delta[1] + \frac{1}{2}h[0] = 0 + \frac{1}{2} = \frac{1}{2}$$

$$h[2] = \delta[2] + \frac{1}{2}h[1] = 0 + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2^2}$$

(b) Extending the solution of (a) and using the unit step function, we may write $h[n]$ as

$$h[n] = \frac{1}{2^n} u[n]$$

(c) A DT LTI system is causal if $h[n] = 0, n < 0$, which is the case here. The system is therefore causal.

(d) A DT LTI system is memoryless if $h[n] = 0, n \neq 0$, which is not that case here. The system is therefore not memoryless.

(e) A DT LTI system is BIBO stable if $\sum_n |h[n]| < \infty$. For the current system, this leads to a truncated geometric series with ratio $r = \frac{1}{2}$, which is a convergent series. The system is therefore stable.

Problem

An LTID system has an impulse response given by

$$h[n] = \left(\frac{1}{2}\right)^{|n|}$$

- (a) Is the system causal? Justify your answer.
- (b) Compute $\sum_{n=-\infty}^{\infty} |h[n]|$. Is this system BIBO-stable?

Solution

(a) No, the system is not causal since $h[n] \neq 0$ for $(n < 0)$.

(b)

$$\begin{aligned}\sum_{n=-\infty}^{\infty} |h[n]| &= \sum_{n=-\infty}^{\infty} \left| \left(\frac{1}{2} \right)^{|n|} \right| = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2} \right)^{|n|} \\ &= \sum_{n=-\infty}^{-1} 0.5^{-n} + \sum_{n=0}^{\infty} 0.5^n = \sum_{n=-\infty}^{-1} 2^n + \sum_{n=0}^{\infty} 0.5^n \\ &= \frac{0 - 2^0}{1 - 2} + \frac{0.5^0 - 0}{1 - 0.5} = 1 + 2 = 3.\end{aligned}$$

Since $h[n]$ is absolutely summable, the system is BIBO stable.

Problem

A DT LTI system has impulse response $h[n]$ and is given an input $x[n]$. Use convolution table to $y[n] = x[n] * h[n]$.

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

$$x[n] = 4^{-n}u[n].$$

$x_1[n]$	$x_2[n]$	$x_1[n] * x_2[n] = x_2[n] * x_1[n]$
$\delta[n - k]$	$x[n]$	$x[n - k]$
$\gamma^n u[n]$	$u[n]$	$\left[\frac{1 - \gamma^{n+1}}{1 - \gamma} \right] u[n]$
$u[n]$	$u[n]$	$(n + 1)u[n]$
$\gamma_1^n u[n]$	$\gamma_2^n u[n]$	$\left[\frac{\gamma_1^{n+1} - \gamma_2^{n+1}}{\gamma_1 - \gamma_2} \right] u[n] \quad \gamma_1 \neq \gamma_2$
$u[n]$	$nu[n]$	$\frac{n(n + 1)}{2} u[n]$
$\gamma^n u[n]$	$nu[n]$	$\left[\frac{\gamma(\gamma^n - 1) + n(1 - \gamma)}{(1 - \gamma)^2} \right] u[n]$
$nu[n]$	$nu[n]$	$\frac{1}{6} n(n - 1)(n + 1)u[n]$

Solution

The input can be expressed as $x[n] = 4^{-n}u[n] = (1/4)^n u[n] = (0.25)^n u[n]$.

$$\begin{aligned}y[n] &= x[n] * h[n] \\&= (0.25)^n u[n] * [(-0.2)^n u[n] + 4(0.8)^n u[n]] \\&= (0.25)^n u[n] * (-0.2)^n u[n] + (0.25)^n u[n] * 4(0.8)^n u[n]\end{aligned}$$

We use pair 4 (Table 3.1) to find the foregoing convolution sums.

$$\begin{aligned}y[n] &= \left[\frac{(0.25)^{n+1} - (-0.2)^{n+1}}{0.25 - (-0.2)} + 4 \frac{(0.25)^{n+1} - (0.8)^{n+1}}{0.25 - 0.8} \right] u[n] \\&= (2.22[(0.25)^{n+1} - (-0.2)^{n+1}] - 7.27[(0.25)^{n+1} - (0.8)^{n+1}])u[n] \\&= [-5.05(0.25)^{n+1} - 2.22(-0.2)^{n+1} + 7.27(0.8)^{n+1}]u[n]\end{aligned}$$

Recognizing that

$$\gamma^{n+1} = \gamma(\gamma)^n$$

we can express $y[n]$ as

$$\begin{aligned}y[n] &= [-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n]u[n] \\&= [-1.26(4)^{-n} + 0.444(-0.2)^n + 5.81(0.8)^n]u[n]\end{aligned}$$

Problem

Determine whether the following systems are BIBO-stable: **(a)** $y(t) = x^2(t)$, **(b)** $y(t) = tx(t)$, and **(c)** $y(t) = \frac{d}{dt}x(t)$.

Solution

(a) This system squares an input to produce the output. If the input is bounded, which is to say that $|x(t)| \leq M_x < \infty$ for all t , then we see that

$$|y(t)| = |x^2(t)| = |x(t)|^2 \leq M_x^2 < \infty$$

Since the output amplitude is guaranteed to be bounded for any bounded-amplitude input, the system $y(t) = x^2(t)$ is BIBO-stable.

(b) We can prove that $y(t) = tx(t)$ is not BIBO-stable with a simple example. The bounded-amplitude input $x(t) = u(t)$ produces the output $y(t) = tu(t)$ whose amplitude grows to infinity as $t \rightarrow \infty$. Thus, $y(t) = tx(t)$ is a BIBO-unstable system.

(c) We can prove that $y(t) = \frac{d}{dt}x(t)$ is not BIBO-stable with an example. The bounded-amplitude input $x(t) = u(t)$ produces the output $y(t) = \delta(t)$ whose amplitude is infinite at $t = 0$. Thus, $y(t) = \frac{d}{dt}x(t)$ is a BIBO-unstable system.

Problem

Determine whether the following systems are memoryless: **(a)** $y(t - 1) = 2x(t - 1)$, **(b)** $y(t) = \frac{d}{dt}x(t)$, and **(c)** $y(t) = (t - 1)x(t)$.

Solution

(a) In this case, the output at time $t - 1$ is just twice the input at the same time $t - 1$. Since the output at a particular time depends only on the strength of the input at the same time, the system is memoryless.

(b) Although it appears that the output $y(t)$ at time t depends on the input $x(t)$ at the same time t , we know that the slope (derivative) of $x(t)$ cannot be determined solely from a single point. There must be some memory, even if infinitesimally small, involved. This is confirmed by using the fundamental theorem of calculus to express the system as

$$y(t) = \lim_{T \rightarrow 0} \frac{x(t) - x(t - T)}{T}$$

Since the output at a particular time depends on more than just the input at the same time, the system is not memoryless.

(c) The output $y(t)$ at time t is just the input $x(t)$ at the same time t multiplied by the (time-dependent) coefficient $t - 1$. Since the output at a particular time depends only on the strength of the input at the same time, the system is memoryless.

Problem

Determine whether the following systems are causal: **(a)** $y(t) = x(-t)$, **(b)** $y(t) = x(t + 1)$, and **(c)** $y(t + 1) = x(t)$.

Solution

(a) System is non-causal. For example while $y(10) = x(-10)$, we will have $y(-10) = x(10)$, so there clearly are past output values that depend on future input values.

(b) In this case, the output at time t depends on the input at future time of $t + 1$. Clearly the system is not causal.

(c) In this case, the output at time $t + 1$ depends on the input one second in the past, at time t . Since the output does not depend on future values of the input, the system is causal.

Problem

$$y(t) = \begin{cases} x(t) & \text{if } x(t) > 0 \\ 0 & \text{if } x(t) \leq 0 \end{cases}$$

Basically, this system clips any negative-amplitude part of the signal.

- (a) Is this system BIBO-stable?
- (b) Is the system causal?
- (c) Is the system invertible?
- (d) Is the system linear?
- (e) Is the system memoryless?
- (f) Is the system time invariant?

Solution

- (a) From the definition of the system, we know that $y(t)$ is either $x(t)$ or 0. Correspondingly, if $|x(t)| < \infty$ then $|y(t)| < \infty$, and the system must be BIBO stable.

Yes. The system is BIBO stable.

- (b) By inspection, it is clear that the system output only depends on the current input. No future (or past) values are involved.

Yes. The system is causal.

- (c) Any negative values of the input are lost and cannot be recovered. Thus, the system is not invertible. For example, $x_1(t) = -1$ and $x_2(t) = -2$ both produce output $y(t) = 0$; it is clearly impossible to always recover an input from its output.

No. The system is not invertible.

(d)

For more involved systems like these, we can also try to prove by counterexamples or by testing additivity and homogeneity separately.

Let's see two approaches.

$$y_1(t) = 0 \text{ at least for } x_1(t) \leq 0$$

$$y_2(t) = 0 \text{ at least for } x_2(t) \leq 0$$

Which clearly means

$$y_1(t) + y_2(t) = 0 \text{ at least for } x_1(t) \leq 0 \text{ and } x_2(t) \leq 0$$

$$x_1(t) \leq -x_2(t)$$

But $y_3(t) = S\{x_1(t) + x_2(t)\} = 0$ at least for $x_1(t) + x_2(t) \leq 0$



Since the two lead to different conditions, we conclude that the system fails additivity and cannot be linear.

Here's an example of proof by counterexample:

Consider two signals: $x_1(t) = 1$ and $x_2(t) = \cos(t)$. The corresponding outputs of these individual signals is $y_1(t) = 1$ and $y_2(t) = \begin{cases} \cos(t) & \text{if } \cos(t) > 0 \\ 0 & \text{if } \cos(t) \leq 0 \end{cases}$. However, if we create a third input $x_3(t) = x_1(t) + x_2(t)$, the system output is $y_3(t) = 1 + \cos(t) \neq y_1(t) + y_2(t)$. Since superposition does not apply, the system cannot be linear.

No. The system is not linear.



Since $1 + \cos(t)$ is always non-negative.

(e) By inspection, it is clear that the system output only depends on the current input. No memory is involved.

Yes. The system is memoryless.

(f) For time-invariance, we have to show that if $y(t) = S\{x(t)\}$ and $y_2(t) = S\{x(t - \tau)\}$, then $y_2(t) = y(t - \tau)$. Let's check:

$$y(t) = \begin{cases} x(t) & \text{if } x(t) > 0 \\ 0 & \text{if } x(t) \leq 0 \end{cases} \quad \longrightarrow \quad y(t - \tau) = \begin{cases} x(t - \tau) & \text{if } x(t - \tau) > 0 \\ 0 & \text{if } x(t - \tau) \leq 0 \end{cases}$$

$$y_2(t) = \begin{cases} x(t - \tau) & \text{if } x(t - \tau) > 0 \\ 0 & \text{if } x(t - \tau) \leq 0 \end{cases}$$

Clearly $y_2(t) = y(t - \tau)$, and we conclude that the system is time-invariant.

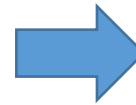
Problem

$$y(t) = \int_t^{t+1} x(\tau - \alpha) d\tau$$

Consider the system above where $x(t)$ and $y(t)$ are the input and output, respectively, and $\alpha < \infty$ is a constant. Determine whether this system is **(a) Memoryless (b) Causal (c) Invertible (d) Time-Invariant, (e) BIBO Stable**.

Provide proper justifications (w.r.t given system) for your answers.

$$y(t) = S\{x(t)\} = \int_t^{t+1} x(\tau - \alpha) d\tau$$



$$y(t - t_0) = \int_{t-t_0}^{t-t_0+1} x(\tau - \alpha) d\tau = \text{area under } x(t) \text{ from } t - t_0 - \alpha \text{ to } t - t_0 + 1 - \alpha$$

$$y_2(t) = S\{x(t - t_0)\} = \int_t^{t+1} x(\tau - t_0 - \alpha) d\tau = \text{area under } x(t) \text{ from } t - t_0 - \alpha \text{ to } t + 1 - t_0 - \alpha$$

Clearly, $y_2(t) = y(t - t_0)$

Solution

We first note that the system represents area under $x(t)$ from $x(t - \alpha)$ to $x(t + 1 - \alpha)$.

(a) System is not memoryless as it requires past and future values (e.g., $x(t + 1 - \alpha)$).

(b) System is, in general, non-causal. However, for $\alpha \geq 1$ the system is causal as in that case all the values from $x(t - \alpha)$ to $x(t + 1 - \alpha)$ represent values before $x(t)$.

(c) System is not invertible, since different functions could give same area.

(d) System is time-invariant as shifting the input by t_0 would also shift the output by the same amount (can be seen by shifting the limits).

$$y(t) = S\{x(t)\} = \int_t^{t+1} x(\tau - \alpha) d\tau \quad \Rightarrow \quad y(t - t_0) = \int_{t-t_0}^{t-t_0+1} x(\tau - \alpha) d\tau = \text{area under } x(t) \text{ from } t - t_0 - \alpha \text{ to } t - t_0 + 1 - \alpha$$

$$y_2(t) = S\{x(t - t_0)\} = \int_t^{t+1} x(\tau - t_0 - \alpha) d\tau = \text{area under } x(t) \text{ from } t - t_0 - \alpha \text{ to } t + 1 - t_0 - \alpha$$

$$\text{Clearly, } y_2(t) = y(t - t_0)$$

(e) System is BIBO stable since the area under a bounded $x(t)$ from $x(t - \alpha)$ to $x(t + 1 - \alpha)$ would be bounded.

Questions?? Thoughts??

